HYBRID PARTICIPATION INNOVATION LAB · STEP 8 · 24 NOVEMBER

Prototype: the Fair Share Simulator

A working tool that lets tenants try fairness rules on real-looking families before they vote

Next Wave

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WHAT IT IS

Make the fairness choice tangible

The problem it solves

"Fair sharing" is abstract. Tenants can't vote on a formula they can't see working.

The simulator shows, family by family, what each rule would mean in euros.

How it works

1. Pick a sunny or cloudy month.

2. Mix the fairness rules with sliders.

3. See each household's share change live, and why.

THE RULES TENANTS CAN MIX

Four ideas of what is fair

Equal

Every household gets the same amount.

Simple, but ignores need and effort.

Need

More for low-income or larger households.

Fights energy poverty, needs sensitive data.

Effort

More for those who use energy in sunny hours.

Raises the bonus, penalises shift workers.

Floor

A minimum guaranteed to everyone, whatever the mix.

Protects those who can't shift.

Tenants set the weight of each rule; the simulator shows the trade-offs. The final mix is decided at Gate 2.

FEATURES

Built to be explained, not just used

Household cards

Six personas: a pensioner, a shift worker, a large family and more

Sun and demand view

A day chart: solar production against each family's use

Rule mixer

Sliders for equal, need, effort and the floor

"Why this amount?"

Each share broken down in plain words

Privacy switch

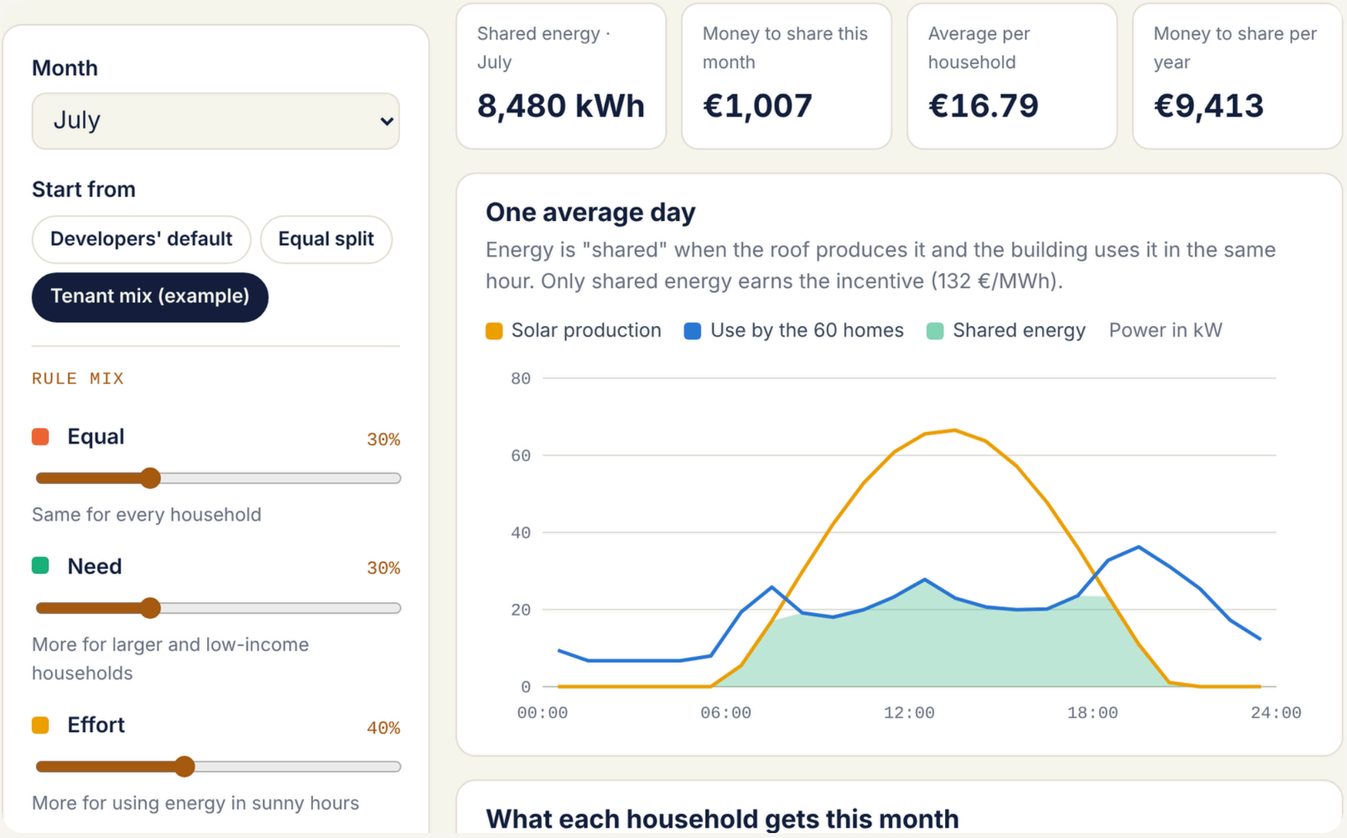
Shows what data each rule needs to work

Paper mode

Printable cards and tables for the table version

DELIVERABLE F · WORKING PROTOTYPE

Try it: the Fair Share Simulator



1 · Pick a month

See solar production, household use and the shared energy that earns the incentive.

2 · Mix the rules

Equal, Need, Effort and a Floor, or a preset: the developers' default or a tenant mix.

3 · Ask why

Every amount is explained in plain words, with the data each rule needs.

kuhyar-saeedi.github.io /next-wave/simulator.html

HOW WE WOULD TEST IT

Three tests with real participants

Test	What participants do	What we measure
Understanding	Explain in their own words why a family got its amount	Share of correct explanations
Usability	Elderly and non-digital tenants use paper or screen with a facilitator	Tasks completed, help needed
Deliberation	The fairness assembly uses it to compare two rules	Did views change? Was the vote more informed?

Before the real process, we can run a first test with classmates playing the household personas.

WHAT TESTING TEACHES US

Questions the prototype answers

Which rule do tenants prefer?

And does the preference change once they see the effect on others?

Is the algorithm explainable?

If tenants can't explain it, the final AI must be simpler.

What data are tenants willing to share?

The answer shapes the data charter (Gate 3).

Does paper work as well as screen?

If not, the hybrid design needs more support.